XISOSIS · Institutional Research Paper

SMOOTHNESS-INVARIANT HANKEL RANK, PRIMITIVE-CHARACTER BOUNDS FOR HIGHER p-ADIC DIGITS, AND QUADRATIC VALUATION-CLOCK ESTIMATION ON Qp-FOLIATIONS

Main result: for every prime p the transfer-matrix Hankel rank of the Qp-foliation phase is exactly p − 1, so smoothness of p − 1 cannot collapse the separation; the m-th p-adic digit has rank between $(p-1)^2$ and $p^{m-1}(p-1)^2$, exactly $(p-1)^2$ for m = 1; and valuation-clock amplitude estimation yields a quadratic speedup on non-uniform networks.

Roger Cooperman
https://orcid.org/0009-0009-5011-2135
and A.E. Petrunina V.V.
https://orcid.org/0009-0002-4595-736X

This white paper — DOI: 10.5281/zenodo.23470489

(Cooperman & Petrunina, 2026Qp-Foliation Interfere DOI: 10.5281/zenodo.23373706

Code and validation suite — GitHub: https://github.com/teknoquants/Hankel-rank-of-the-Qp-foliation-phase-

Vienna · 18 October 2026

© 2026 Roger Cooperman; A.E. Petrunina V.V. · Released under the MIT License (see Licence)

Abstract. This white paper resolves three structural boundaries of the Qp-Foliation Interference Theorem identified in the open problems of Cooperman & Petrunina (2026; DOI: 10.XXXX/XXXXX — Qp-Foliation Interference Theorem). The three boundaries are smooth-order fields, higher-order p-adic digits, and non-uniform valuation clocks. For each we prove a sharp statement in the transfer-matrix model and pair it with a reproducible numerical validation.

Results. (1) In the transfer-matrix model the separation is *invariant* under smoothness: the Hankel rank of $\psi \circ \bar{a}$ is exactly $p - 1$ for every prime p, and smoothness only makes the quotient-sector sums (characters of order dividing Q) cheap. (2) In the pure-carry regime $m < t_{min}$, the m-th digit phase has Hankel rank at least $\rho_m(p)$. We prove $\rho_1(p) = (p-1)^2$, $\rho_m(p) \le p^{m-1}(p-1)^2$ and $\rho_m(p) \ge (p-1)^2$, and conjecture equality, which holds for all $p \le 31$, $m \le 3$. (3) A valuation-clock estimator returns $S(L_m)/N$ to additive error ε with $O(1/\varepsilon)$ oracle calls, a quadratic gain over classical sampling. Every statement is paired with an MIT-licensed Python validation script; the suite runs 214 checks, all passing.

1. Setting, hypotheses and computational model

1.1 Combinatorial data

Let X be a finite graph with set E of oriented edges, $d = |E|$. A transition $e \to e'$ is *non-backtracking* (NB) if the head of e is the tail of e' and e' is not the reverse of e. A *closed NB word* of length n is a rooted sequence $\delta = (e_1, \ldots, e_n)$ in which every cyclic transition $e_k \to e_{k+1}$ (indices mod n, so including $e_n \to e_1$) is NB.

Fix an odd prime p, the additive character $\psi(x) = \exp(2\pi i\, x/p)$ on $\mathbb{F}_p$, and junction digits $a(e, e') \in \mathbb{F}_p^\times$ on NB transitions. For a closed word put $\bar{a}(\delta) = \prod_k a(e_k, e_{k+1})$ (cyclic), and

$$S_n = \sum_{|\delta| = n,\ \delta \text{ closed NB}} \psi(\bar{a}(\delta)).$$

For a multiplicative character χ of $\mathbb{F}_p^\times$ let $(B_\chi)_{e,e'} = \chi(a(e, e'))$ on NB transitions and 0 otherwise, and let $G(\chi) = \sum_{t \ne 0} \chi(t)\psi(t)$ be the Gauss sum; χ_0 is the trivial character.

1.2 Standing hypotheses

Axioms III′–IV and Hypotheses (H), (H_m) are restated here in the form used below. Axiom III′ makes the spectral normalisation explicit; Lemma 4.1(c) shows it cannot be dropped.

- **Axiom III′ (spectral contraction).** Each edge carries $\nabla_e = u_e v_e^{T} + \mu(e) R_e \in M_2(\mathbb{Z}_p)$ with $u_e, v_e \in \mathbb{Z}_p^2$, $v_e^{T} u_e = 1$ and $R_e = I - u_e v_e^{T}$. Thus u_e is an eigenvector with eigenvalue 1 and $\ker v_e^{T}$ is the eigenline for $\mu(e)$.
- **Axiom IV (contraction exponents).** $\mu(e) \in p^{t_e}\mathbb{Z}_p^\times$ with $t_e \ge 1$, and the junction weights $w(e, e') := v_{e'}^{T} u_e$ are p-adic units on every NB transition. Write $t_{min} = \min t_e$, $t_{max} = \max t_e$.
- **Hypothesis (H).** There are edges e_0, e_a, e_b, e_1 with $e_a \to e_b$ and $e_1 \to e_0$ NB such that the open products $X(u) = \prod a$ along NB words u from e_0 to e_a, and $Y(v)$ along NB words v from e_b to e_1, each cover $\mathbb{F}_p^\times$.
- **Hypothesis (H_m).** The same with a replaced by w and products taken in $G_m := (\mathbb{Z}/p^{m+1})^\times$, which they must cover. (H_0) is (H) for the leading digits $a = w \bmod p$.

For a closed word δ write $\gamma_\delta = \nabla_{e_n} \cdots \nabla_{e_1}$, $W(\delta) = \prod_k w(e_k, e_{k+1})$ (cyclic), $T(\delta) = \sum_k t_{e_k}$, and $r(\delta) = \sum_{j \ge 0} \alpha_j(\delta) p^j$ for the unit eigenvalue of γ_δ (Lemma 3.1(b)). $\mathrm{digit}_m(x)$ denotes the m-th base-p digit of $x \in \mathbb{Z}_p$. When $a = w \bmod p$, Lemma 3.1(c) with $m = 0$ gives $\alpha_0(\delta) = \bar{a}(\delta)$, which links the two descriptions.

1.3 Computational model

A *transfer-matrix representation* of a word function φ is a weighted automaton over ℂ: vectors λ, κ and matrices M_e with $\varphi(e_1 \cdots e_n) = \lambda^T M_{e_1} \cdots M_{e_n} \kappa$. By the Fliess / Carlyle–Paz theorem, the minimal dimension equals the rank of the Hankel matrix $H_\varphi[u, v] = \varphi(uv)$. The lower bounds proved here are lower bounds *in this model only*. They do not exclude classical algorithms of other kinds, and the closing remark of §2 returns to this point.

1.4 Validation

Each statement carries a tag naming the script in `validation/` that checks it numerically or exactly. Scripts use exact integer arithmetic in $\mathbb{Z}/p^K$ for p-adic statements and NumPy for spectral and rank computations. The test core is $X = K_4$ ($d = 12$, two NB successors per edge). Section 6 lists the scripts and results.

2. Extension 1: smooth-order fields

> **Lemma 2.1 (Pohlig–Hellman character reduction).** *Let $p - 1 = \prod_i q_i^{e_i}$ with every $q_i \le B$, and let g generate $\mathbb{F}_p^\times$. For $y \in \mathbb{F}_p^\times$ the discrete logarithm $x = \log_g y$, and hence $\chi(y) = \chi(g)^x$ for every multiplicative character χ, can be computed with $O(\sum_i e_i(\log p + q_i))$ multiplications in $\mathbb{F}_p$. Since $\sum e_i \le \log_2 p$, this is $O(\log p \cdot (\log p + B))$.*
>
> Validated by `v01_lemma_2_1_pohlig_hellman.py` (30 checks; measured count ≤ $1.4 \cdot \sum e_i(\log_2 p + q_i)$)

Proof. By the Chinese remainder theorem, x is determined by its residues x mod q^e for each prime power $q^e \parallel p - 1$. Fix such a q^e and write $x \bmod q^e = \sum_{j<e} x_j q^j$. Put $s_j = x_0 + x_1 q + \cdots + x_{j-1} q^{j-1}$. With $\gamma = g^{(p-1)/q}$ of order q, raising $y \cdot g^{-s_j}$ to the power $(p-1)/q^{j+1}$ gives γ^{x_j}. Computing this power costs O(log p) multiplications by square-and-multiply. Recovering x_j by exhaustive search in $\langle\gamma\rangle$ costs at most q multiplications. Summing over j and over the prime powers gives the bound, and CRT recombination is negligible. Finally $\chi(y) = \chi(g)^x = \exp(2\pi i\, kx/(p-1))$ when $\chi(g) = \exp(2\pi i\, k/(p-1))$. ■

> **Theorem 2.2 (Smoothness invariance of the sector bottleneck).** *Let p be any odd prime, with no condition on $p - 1$. Then:*
>
> *(a) $S_n = \sum_\chi G(\bar\chi)/(p-1) \cdot \mathrm{Tr}(B_\chi^n)$, the sum running over all $p - 1$ multiplicative characters.*
>
> *(b) $G(\chi_0) = -1$ and $|G(\chi)| = \sqrt{p}$ for every $\chi \ne \chi_0$; no sector has vanishing weight.*
>
> *(c) Under Hypothesis (H), the Hankel matrix of $\varphi = \psi \circ \bar a$ restricted to closed words has rank exactly $p - 1$ on the prefix/suffix block of (H). Every transfer-matrix representation of φ therefore has dimension $\ge p - 1$, whatever the factorisation of $p - 1$.*
>
> *(d) Consequently the sector-by-sector algorithm, which is the block-diagonal transfer matrix $\bigoplus_\chi B_\chi$ of dimension $d(p-1)$, uses $\Theta(p \cdot d^3 \log n)$ arithmetic operations plus the entry cost of Lemma 2.1, even when $p - 1$ is B-smooth. Smoothness does not collapse the separation in this model.*
>
> Validated by `v02_theorem_2_2_smoothness_invariance.py` (26 checks; smooth $p \in \{17, 97\}$, non-smooth $p \in \{23, 47, 59\}$)

Proof. (a) Orthogonality gives $\sum_\chi \chi(x)\bar\chi(t) = (p-1) \cdot 1[x = t]$ on $\mathbb{F}_p^\times$, hence $\psi(x) = (p-1)^{-1} \sum_\chi G(\bar\chi)\chi(x)$. Substituting $x = \bar a(\delta)$ and using $\chi(\bar a(\delta)) = \prod_k \chi(a(e_k, e_{k+1}))$, the sum of $\chi(\bar a(\delta))$ over closed NB words of length n equals $\mathrm{Tr}(B_\chi^n)$.

(b) $G(\chi_0) = \sum_{t \ne 0} \psi(t) = -1$. For $\chi \ne \chi_0$, $|G(\chi)|^2 = p$ is classical (Ireland–Rosen, Ch. 8).

(c) For u from e_0 to e_a and v from e_b to e_1, the concatenation uv is a closed NB word and $\bar a(uv) = c \cdot X(u)Y(v)$ with $c = a(e_a, e_b)a(e_1, e_0) \in \mathbb{F}_p^\times$. Rows with the same X(u) are equal, so by (H) the block has the rank of the group matrix $M[X, Y] = f(XY)$, $X, Y \in \mathbb{F}_p^\times$, where $f(Z) = \psi(cZ)$. Characters diagonalise group matrices: $f(XY) = (p-1)^{-1} \sum_\chi \hat f(\chi)\chi(X)\chi(Y)$ with $\hat f(\chi) = \sum_Z f(Z)\bar\chi(Z) = \chi(c)G(\bar\chi)$. By (b) every $\hat f(\chi)$ is

nonzero, and the characters are linearly independent, so rank M = p − 1. Since H_φ contains this block, Carlyle–Paz gives the bound. No property of p − 1 was used.

(d) $\bigoplus_\chi B_\chi$ is itself a transfer matrix computing S_n, of dimension $d(p-1) \geq p-1$, consistent with (c). Evaluating each $\mathrm{Tr}(B_\chi^n)$ by repeated squaring costs $O(d^3 \log n)$, and there are p − 1 of them. ■

Remark (limits of smoothness). B-smoothness gives only $p - 1 \leq B^{\Sigma e_i}$, which does not make p − 1 polynomial in B or log p: with $\sum e_i$ up to $\log_2 p$ it permits p − 1 itself. The Davenport–Hasse relation links Gauss sums across extensions of $\mathbb{F}_p$ and does not evaluate $G(\chi)$ over $\mathbb{F}_p$ in poly(log p) time. Stickelberger's theorem fixes the ideal generated by $G(\chi)^{\mathrm{ord}\,\chi}$ but leaves a unit undetermined.

Proposition 2.3 (Quotient sectors: what smoothness buys). *Let $Q \mid p-1$, $\zeta = g^{(p-1)/Q}$, and $\lambda_Q(x) = \log_\zeta(x^{(p-1)/Q}) \in \mathbb{Z}/Q$, a homomorphism $\mathbb{F}_p^\times \to \mathbb{Z}/Q$ with kernel $H_Q = (\mathbb{F}_p^\times)^Q$. Let $\Xi_Q = \{\chi : \chi^Q = 1\}$ and define $S_n^{[Q]} = \sum_{\chi \in \Xi} G(\bar\chi)/(p-1) \cdot \mathrm{Tr}(B_\chi^n)$, the sum over $\chi \in \Xi_Q$. Then*

$$S_n^{[Q]} = \sum_{c \in \mathbb{Z}/Q} N_n(c) \cdot \eta_Q(c), \quad \eta_Q(c) = (Q/(p-1)) \cdot \sum \{\psi(t) : \lambda_Q(t) = c\},$$

where $N_n(c)$ counts closed NB words of length n with $\lambda_Q(\bar a(\delta)) = c$. N_n is produced by a transfer matrix of dimension dQ on $E \times \mathbb{Z}/Q$. Its entries $\lambda_Q(a(e, e'))$ cost $O(\log p + \sum_{q^r \| Q} f(q + \log Q))$ multiplications each, by Pohlig–Hellman inside μ_Q. The walk-dependent cost is therefore poly(n, d, Q, B, log p) when Q is B-smooth. The Q Gaussian periods η_Q are a walk-independent precomputation. For $Q = p - 1$ the identity reduces to S_n.

Validated by `v03_proposition_2_3_quotient_sectors.py` (53 checks; p = 97 and p = 7681, Q ∈ {2, …, 96})

Proof. $\chi \in \Xi_Q$ exactly when χ is trivial on H_Q, so these Q characters are the characters of $\mathbb{F}_p^\times / H_Q \cong \mathbb{Z}/Q$. Then, with all character sums over $\chi \in \Xi_Q$, $(p-1)^{-1}\sum_\chi G(\bar\chi)\chi(x) = (p-1)^{-1}\sum_t \psi(t)\sum_\chi \chi(xt^{-1}) = (Q/(p-1)) \cdot \sum\{\psi(t) : t \in xH_Q\} = \eta_Q(\lambda_Q(x))$. Summing over closed words and grouping by $\lambda_Q(\bar a(\delta)) = \sum_k \lambda_Q(a(e_k, e_{k+1}))$ mod Q gives the identity. $N_n(c) = \sum_e (T^n)_{(e,0),(e,c)}$ for the 0/1 transfer matrix T that moves state (e, c) to $(e', c + \lambda_Q(a(e, e')))$ along each NB transition. The cost of λ_Q is one exponentiation followed by Lemma 2.1 inside the cyclic group μ_Q of order Q. ■

Remark (scope). Theorem 2.2(c) is a lower bound for transfer-matrix representations, the model in which the original separation is stated. It is not a lower bound against arbitrary classical algorithms, and a poly(log p) classical algorithm for S_n outside this model is not excluded by anything proved here.

3. Extension 2: higher-order digits and carries

Lemma 3.1 (Higher-digit expansion). *Assume Axioms III′–IV and let $\delta = (e_1, \ldots, e_n)$ be a closed NB word. For $\sigma \subseteq \{1, \ldots, n\}$ put $\Phi_\sigma(\delta) = \mu_\sigma \cdot \mathrm{Tr}(M_n^\sigma \cdots M_1^\sigma)$, where $M_k^\sigma = R_{e_k}$ if $k \in \sigma$ and $u_{e_k} v_{e_k}^T$ otherwise, $\mu_\sigma = \prod_{k \in \sigma} \mu(e_k)$ and $t(\sigma) = \sum_{k \in \sigma} t_{e_k}$. Then:*

(a) $\mathrm{Tr}\,\gamma_\delta = \sum_\sigma \Phi_\sigma(\delta)$, with $\Phi_\emptyset(\delta) = W(\delta)$ and $\mathrm{val}_p\,\Phi_\sigma(\delta) \geq t(\sigma)$.

(b) $\det \gamma_\delta = \prod_k \mu(e_k)$ has valuation $T(\delta)$. γ_δ has a unique unit eigenvalue $r(\delta)$, with $r(\delta) \equiv \mathrm{Tr}\,\gamma_\delta \pmod{p^{T(\delta)}}$ and $\mathrm{val}_p(r(\delta) - \mathrm{Tr}\,\gamma_\delta) = T(\delta)$ exactly.

(c) For every $m < T(\delta)$: $\alpha_m(\delta) = \mathrm{digit}_m\big(\sum_{t(\sigma) \leq m} \Phi_\sigma(\delta) \bmod p^{m+1}\big)$. In particular, if $m < t_{\min}$ then $\alpha_m(\delta) = \mathrm{digit}_m(W(\delta))$.

(d) Carries are multiplicative: for units x, y, $\mathrm{digit}_1(xy) \equiv x_0 y_1 + x_1 y_0 + \lfloor x_0 y_0 / p \rfloor \pmod p$, and in general $\mathrm{digit}_m(xy)$ is a polynomial in the digits 0, …, m of x and y. For $m \geq 1$ it is not a function of $(\mathrm{digit}_m x, \mathrm{digit}_m y)$.

Validated by `v04_lemma_3_1_carry_expansion.py` (21 checks; p ∈ {3, 5, 7}, $t_e \in \{1, 2, 3\}$, n ≤ 6; also confirms the cut-off $t(\sigma) \leq m$ is sharp)

Proof. (a) Expand $\gamma_\delta = \prod_k (u_{e_k} v_{e_k}^T + \mu(e_k) R_{e_k})$ over subsets σ of positions taking the R-term. All entries are integral, so val $\Phi_\sigma \geq$ val $\mu_\sigma = t(\sigma)$. For $\sigma = \emptyset$ the product of rank-one factors telescopes: $u_{e_n} v_{e_n}^T \cdots u_{e_1} v_{e_1}^T = (\prod_{k<n} v_{e_{k+1}}^T u_{e_k})\, u_{e_n} v_{e_1}^T$, whose trace adds the closing factor $v_{e_1}^T u_{e_n}$, giving $W(\delta)$.

(b) $\det \nabla_e = \mu(e)$ by Lemma 4.1(b), and det is multiplicative. By (a), $\operatorname{Tr} \gamma_\delta \equiv W(\delta) \pmod{p^{t_{min}}}$, a unit. The characteristic polynomial $x^2 - (\operatorname{Tr} \gamma_\delta)x + \det \gamma_\delta$ reduces mod p to $x(x - \operatorname{Tr} \gamma_\delta)$, which has simple roots. By Hensel's lemma there is a unique unit root $r(\delta)$, and the other root $\det \gamma_\delta / r(\delta)$ has valuation $T(\delta)$. From $r = \operatorname{Tr} \gamma_\delta - \det \gamma_\delta / r$ we get $\operatorname{val}(r - \operatorname{Tr} \gamma_\delta) = T(\delta)$.

(c) If $m < T(\delta)$, then $r \equiv \operatorname{Tr} \gamma_\delta \pmod{p^{m+1}}$ by (b), and by (a) only terms with $t(\sigma) \leq m$ survive modulo p^{m+1}. If $m < t_{min}$, every nonempty σ has $t(\sigma) \geq t_{min} > m$, leaving $\Phi_\emptyset = W(\delta)$.

(d) Write $x = x_0 + p x_1 + \cdots$ and y likewise. Then $xy \equiv x_0 y_0 + p(x_0 y_1 + x_1 y_0) \pmod{p^2}$, and the carry $\lfloor x_0 y_0 / p \rfloor$ enters digit 1. The validation finds pairs with equal (x_1, y_1) but different $\text{digit}_1(xy)$. ■

Remark (source of carries). Two points in Lemma 3.1 deserve emphasis. The m-th digit is extracted from a residue modulo p^{m+1}, and the cut-off is $t(\sigma) \leq m$ rather than a bound on the number of R-factors. In the regime $m < t_{min}$ the R-terms vanish identically, and every carry comes from multiplying junction weights. For $m \geq t_{min}$ both sources contribute, and the validation confirms that the R-terms then change α_m.

> **Theorem 3.2 (Higher-digit Hankel rank).** *Let $G_m = (\mathbb{Z}/p^{m+1})^\times$, $f_m(X) = \psi(\text{digit}_m X)$, $\hat{f}_m(\chi) = \sum_{X \in G_m} f_m(X) \bar{\chi}(X)$, and $\rho_m(p) = \#\{\chi \in \hat{G}_m : \hat{f}_m(\chi) \neq 0\}$. Assume Axioms III′–IV, $0 \leq m < t_{min}$ and (H_m). Then the Hankel matrix of $\varphi_m = \psi \circ \alpha_m$ on closed words has rank $\geq \rho_m(p)$, so every transfer-matrix representation of φ_m has dimension $\geq \rho_m(p)$. Moreover:*
>
> *(a) $\rho_0(p) = p - 1$.*
>
> *(b) For $m \geq 1$, $\hat{f}_m(\chi) = 0$ for every imprimitive χ (χ trivial on $1 + p^m\mathbb{Z}$), so $\rho_m(p) \leq p^{m-1}(p-1)^2$.*
>
> *(c) For $m \geq 1$ and χ primitive, with $\chi(1 + p^m s) = \exp(2\pi i\, c_\chi s / p)$: $\hat{f}_m(\chi) = p \cdot \sum_{0 \leq x < p^m,\, x \equiv c_\chi \pmod p} \bar{\chi}(x)$. For $m = 1$ this has modulus p, so $\rho_1(p) = (p-1)^2$.*
>
> *(d) For $m \geq 1$, $\rho_m(p) \geq (p-1)^2$.*
>
> Validated by `v05_theorem_3_2_higher_digit_rank.py` (57 checks, including end-to-end Hankel ranks from true eigenvalue digits on K_4)

Proof. Reduction. For u from e_0 to e_a and v from e_b to e_1, uv is closed and $W(uv) = c \cdot X(u) Y(v)$ with $c = w(e_a, e_b) w(e_1, e_0) \in \mathbb{Z}_p^\times$. By Lemma 3.1(c), since $m < t_{min}$, $\varphi_m(uv) = f_m(cX(u)Y(v))$, which depends on X, Y only modulo p^{m+1}. By (H_m) the Hankel matrix contains the group matrix $[f_m(cXY)]_{X,Y \in G_m}$ up to repeated rows and columns. Its rank is the number of characters with $\hat{f}_m(\chi)\chi(c) \neq 0$, which is $\rho_m(p)$. Carlyle–Paz completes the bound.

(a) For $m = 0$, $\hat{f}_0(\chi) = G(\bar{\chi}) \neq 0$ for all χ (Theorem 2.2(b)).

(c) and (b). Write each $X \in G_m$ uniquely as $X = x + p^m d$ with $0 \leq x < p^m$, $p \nmid x$, $0 \leq d < p$. Then $X \equiv x(1 + p^m d\, \bar{x}) \pmod{p^{m+1}}$, where $\bar{x}$ is the inverse of x mod p, and $f_m(X) = \psi(d)$. If χ is trivial on $1 + p^m\mathbb{Z}$, then $\hat{f}_m(\chi) = \sum_x \bar{\chi}(x) \sum_d \psi(d) = 0$, which proves (b). The count of primitive characters is $|G_m| - |G_{m-1}| = p^{m-1}(p-1)^2$. If χ is primitive, $\bar{\chi}(X) = \bar{\chi}(x) \exp(-2\pi i\, c_\chi d \bar{x} / p)$, and the inner sum $\sum_d \exp(2\pi i\, d(1 - c_\chi \bar{x})/p)$ equals p when $x \equiv c_\chi \pmod p$ and 0 otherwise, which gives (c). For $m = 1$ exactly one x qualifies, so $|\hat{f}_1(\chi)| = p$.

(d) Parseval gives $\sum_\chi |\hat{f}_m(\chi)|^2 = |G_m|^2 = p^{2m}(p-1)^2$. By (c), $|\hat{f}_m(\chi)| \leq p \cdot p^{m-1}$, so at least $(p-1)^2$ coefficients are nonzero. ■

> **Conjecture 3.3.** *$\rho_m(p) = p^{m-1}(p-1)^2$ for every odd prime p and every $m \geq 1$. Equivalently, the incomplete character sum in Theorem 3.2(c) never vanishes for primitive χ.*

Validated by `v05_theorem_3_2_higher_digit_rank.py` (verified by FFT over G_m for all $p \le 31$, $m \le 3$, up to $|G_m| \approx 9 \cdot 10^5$)

Corollary 3.4. *Under the hypotheses of Theorem 3.2, every transfer-matrix representation of φ_1 has dimension $\ge (p-1)^2$. For $m \ge 2$ the dimension is $\ge (p-1)^2$ unconditionally, and $\ge p^{m-1}(p-1)^2 = \Omega(p^{m+1})$ assuming Conjecture 3.3. The trace-form transfer matrix on $E \times G_m$, of dimension $d \cdot p^m(p-1)$, computes $\sum_{|\delta|=n} \psi(\alpha_m(\delta))$, so the bound is tight up to the factor $dp/(p-1)$.*

Remark (interpretation). This is the m = 0 separation with modulus p^{m+1} in place of p. A quantum digit register then needs $\lceil (m+1)\log_2 p \rceil$ qubits, so the gap has the same form as before rather than being qualitatively stronger. This paper gives no quantum algorithm for the m-th digit sum. The rank is strictly below the naive count $p^{m+1} - 1$: for p = 5, m = 1 the rank is 16 rather than 24.

4. Extension 3: non-uniform networks and valuation clocks

Lemma 4.1 (Valuation additivity). *(a) For any 2×2 matrices over $\mathbb{Z}_p$ of the form $\nabla_e = u_e v_e^T + \mu(e) R_e$: $\det \nabla_e = \mu(e) \cdot \mathrm{tr}(\mathrm{adj}(u_e v_e^T) R_e) + \mu(e)^2 \det R_e$. Hence $\mathrm{val} \det \nabla_e \ge t_e$, with equality iff the factor $\mathrm{tr}(\mathrm{adj}(u_e v_e^T) R_e) + \mu(e) \det R_e$ is a unit.*

(b) Under Axiom III′, $\det \nabla_e = \mu(e)$. Consequently $\det \gamma_\delta = \prod_k \mu(e_k)$ and $\mathrm{val} \det \gamma_\delta = \sum_k t_{e_k}$ for every walk δ, with t_e allowed to vary from edge to edge.

(c) Without the normalisation, (b) can fail: there are integral R_e with $\mathrm{val} \det \nabla_e > t_e$.

Validated by `v06_lemma_4_1_valuation_additivity.py` (16 checks; explicit counterexample for (c) for each $p \in \{3, 5, 7, 11\}$)

Proof. (a) For 2×2 matrices $\det(A + B) = \det A + \det B + \mathrm{tr}(\mathrm{adj}(A)B)$. Apply this with $A = u_e v_e^T$ ($\det A = 0$) and $B = \mu(e) R_e$. (b) $\nabla_e u_e = u_e (v_e^T u_e) + \mu(e)(u_e - u_e (v_e^T u_e)) = u_e$. For $z \in \ker v_e^T$, $\nabla_e z = \mu(e) z$. Since $v_e^T u_e = 1$, $u_e \notin \ker v_e^T$, so the eigenvalues are 1 and $\mu(e)$ and $\det \nabla_e = \mu(e)$. Multiplicativity of det and additivity of val complete (b). (c) Choose R_e with $\mathrm{tr}(\mathrm{adj}(u_e v_e^T) R_e) \equiv 0 \pmod p$; the validation exhibits one for each tested p. ■

Theorem 4.2 (Valuation-clock leaf estimator). *Assume Axioms III′–IV with non-uniform t_e. Fix a leaf target m and put $k_{max} = \lfloor m / t_{min} \rfloor$. Let P be the set of all NB words of length $1, \dots, k_{max}$, $N = |P|$, and for $\delta \in P$ let $f(\delta) = 1$ iff δ is closed and $T(\delta) = m$. Set $L_m = \{f = 1\}$ and $S(L_m) = \sum_{\delta \in L_m} \psi(\bar{a}(\delta))$. Then:*

(a) A state-preparation unitary A with $A|0\rangle = |\Psi\rangle = N^{-1/2} \sum_{\delta \in P} |\delta\rangle$ can be built from $k_{max} + 1$ layers of controlled rotations, whose angles come from a classical table of NB-extension counts of size $d \cdot k_{max}$.

(b) Let $\tilde{U} = V^\dagger U_\varphi V$, where V computes the clock, closure, digit and flag registers and U_φ applies the phase $\psi(y)$ iff $f = 1$. Then $\langle \Psi | \tilde{U} | \Psi \rangle = (S(L_m) + N - |L_m|)/N$ and $P(f = 1) = |L_m|/N$, so $S(L_m)/N = \langle \tilde{U} \rangle - 1 + P(f = 1)$.

(c) Amplitude estimation on the two Hadamard-test amplitudes and on the flag amplitude, each with M calls and the median of three runs, returns an estimate within $5(\pi/M + \pi^2/M^2)$ of $S(L_m)/N$ with probability $\ge 2/3$. Taking $M = \lceil 10\pi/\varepsilon \rceil$ gives additive error ε with $O(1/\varepsilon)$ calls to A, $\tilde{U}$ and their inverses.

(d) The resulting error on $S(L_m)$ itself is εN, and N grows exponentially in k_{max}. Classical sampling from P needs $\Theta(1/\varepsilon^2)$ samples for the same guarantee, so the improvement is quadratic.

(e) The clock register needs $b = \lceil \log_2(k_{max} t_{max} + 1) \rceil$ qubits, or a saturating adder. With $\lceil \log_2 m \rceil$ qubits the register wraps around and walks with $T(\delta) \equiv m \pmod{2^b}$ but $T(\delta) \ne m$ are flagged.

Validated by `v07_theorem_4_2_valuation_clock.py` (11 checks: statevector Hadamard test, exact AE outcome distributions, error scaling $M^{-1.02}$ vs $M^{-0.51}$, clock wrap-around counterexample)

Circuit. The registers are: path $|\ell, e_1, \dots, e_{k_{max}}\rangle$ ($\lceil \log_2 k_{max} \rceil + k_{max} \lceil \log_2 d \rceil$ qubits), clock $|c\rangle$ (b qubits), digit $|y\rangle$ ($\lceil \log_2 p \rceil$ qubits), closure bit κ, flag f, and one Hadamard ancilla. V performs the following steps in order:

1. **Prepare** $|\Psi\rangle$ with A. First rotate the length register with weights N_ℓ/N. Then choose e_1 with weight $c_\ell(e_1)/N_\ell$ and each subsequent e_i with weight $c_{\ell-i+1}(e_i)/c_{\ell-i+2}(e_{i-1})$, where $c_r(e)$ is the number of NB words of length r starting at e. The product of these weights telescopes to $1/N$.
2. **Clock:** for $i \le \ell$, add t_{e_i} to c with a controlled in-place adder. By Lemma 4.1 the result is $T(\delta)$.
3. **Closure:** $\kappa \leftarrow [e_\ell \to e_1 \text{ is NB}]$.
4. **Digit transport:** $y \leftarrow 1$, then $y \leftarrow y \cdot a(e_i, e_{i+1}) \bmod p$ for $i < \ell$, and the closing junction when $\kappa = 1$. All digits are units, so each step is reversible.
5. **Flag:** $f \leftarrow \kappa \wedge [c = m]$.

U_φ is the diagonal phase $\psi(y)^f$. $\tilde{U} = V^\dagger U_\varphi V$ is used in a Hadamard test, once with and once without an $S^\dagger$ gate on the ancilla, to obtain the real and imaginary parts of $\langle\tilde{U}\rangle$.

Proof. (a) is the telescoping in step 1. (b) $\tilde{U}$ acts on $|\delta\rangle$ as multiplication by $\psi(\bar{a}(\delta))^{f(\delta)}$. Hence $\langle\tilde{U}\rangle = N^{-1}(\sum_{\delta\in L_m}\psi(\bar{a}(\delta)) + |P \setminus L_m|)$. Note that the phase step does *not* zero out paths outside the leaf. Each such path contributes +1, so the correction term $P(f = 1)$ is required.

(c) The Hadamard test returns 0 with probability $a_{Re} = (1 + \mathrm{Re}\langle\tilde{U}\rangle)/2$, or $a_{Im} = (1 + \mathrm{Im}\langle\tilde{U}\rangle)/2$ with $S^\dagger$. Canonical amplitude estimation with M calls returns $\tilde{a}$ with $|\tilde{a} - a| \le 2\pi\sqrt{(a(1-a))/M + \pi^2/M^2} \le \pi/M + \pi^2/M^2$ with probability $\ge 8/\pi^2$ (Brassard, Høyer, Mosca and Tapp 2002). The median of three independent runs lies in this interval with probability ≥ 0.90, so all three components do with probability $\ge 0.74 > 2/3$. The estimate $\hat{S}/N = (2\tilde{a}_{Re} - 1) + i(2\tilde{a}_{Im} - 1) - 1 + \tilde{a}_f$ has error $\le 2e_{Re} + 2e_{Im} + e_f \le 5(\pi/M + \pi^2/M^2)$.

(d) Multiplying by N gives the error on $S(L_m)$. The classical estimator $N^{-1}\sum f\cdot\psi$ over uniform samples is unbiased with variance $\Theta(1)$ in general, so it needs $\Theta(1/\varepsilon^2)$ samples. (e) The clock takes values up to $k_{max}t_{max}$. The validation exhibits false positives when $b = \lceil\log_2 m\rceil$. ■

Remark (design choices). Closure is enforced by the flag, and the superposition runs over all NB words, which is efficiently preparable. A uniform superposition over closed walks is not needed. A clock of $\lceil\log_2 m\rceil$ qubits would not suffice, since it cannot hold m when m is a power of two. The gain is quadratic. If the core theorem's speedup is exponential, this estimator does not inherit it.

5. Conclusion

The three frontiers resolve as follows:

- **Smooth $p - 1$:** within the transfer-matrix model the separation is robust, since the Hankel rank $p - 1$ does not depend on the factorisation of $p - 1$. Smoothness makes quotient-sector sums cheap (Proposition 2.3) but does not reduce the number of sectors.
- **Higher digits:** in the pure-carry regime $m < t_{min}$, the m-th digit phase has Hankel rank $\rho_m(p)$. This equals $(p - 1)^2$ exactly for $m = 1$ and is at least $(p - 1)^2$ for every m. It equals $p^{m-1}(p - 1)^2 = \Omega(p^{m+1})$ under Conjecture 3.3, which holds numerically for all $p \le 31$, $m \le 3$.
- **Valuation clocks:** a valuation-clock circuit estimates the normalised leaf sum $S(L_m)/N$ with $O(1/\varepsilon)$ calls, a quadratic gain over classical sampling.

The open items are a proof of Conjecture 3.3, which amounts to non-vanishing of incomplete character sums modulo p^{m+1}, and a quantum algorithm for the m-th digit sum. Both determine whether Extension 2 strengthens the separation beyond a change of modulus.

6. Reproducibility

The validation suite is available at https://github.com/<OWNER>/<REPOSITORY>. It is MIT-licensed and requires only Python ≥ 3.10 and NumPy. To reproduce every check, run `pip install -r`

`requirements.txt` followed by `python validation/run_all.py`. The suite completes in under 15 seconds on a laptop, and the exit status is non-zero if any check fails.

Statement	Script	Checks	Result
Lemma 2.1	v01_lemma_2_1_pohlig_hellman.py	30	30/30 pass
Theorem 2.2	v02_theorem_2_2_smoothness_invariance.py	26	26/26 pass
Proposition 2.3	v03_proposition_2_3_quotient_sectors.py	53	53/53 pass
Lemma 3.1	v04_lemma_3_1_carry_expansion.py	21	21/21 pass
Theorem 3.2, Conj. 3.3	v05_theorem_3_2_higher_digit_rank.py	57	57/57 pass
Lemma 4.1	v06_lemma_4_1_valuation_additivity.py	16	16/16 pass
Theorem 4.2	v07_theorem_4_2_valuation_clock.py	11	11/11 pass
Total	validation/run_all.py	**214**	**214/214 pass**

Numerical checks are evidence, not proofs. The proofs in §§2–4 carry the logical weight, while the scripts guard against algebraic slips and confirm that every hypothesis is satisfiable and every counterexample real.

How to cite

Cooperman, R., Petrunina, A. E. (2026). Smoothness-Invariant Hankel Rank, Primitive-Character Bounds for Higher p-adic Digits, and Quadratic Valuation-Clock Estimation on Qp-Foliations. White paper. DOI: 10.XXXX/XXXXX. Code: https://github.com/<OWNER>/<REPOSITORY>.

References

Brassard, G., Høyer, P., Mosca, M., Tapp, A. (2002). Quantum amplitude amplification and estimation. *Contemporary Mathematics* 305, 53–74.

Carlyle, J. W., Paz, A. (1971). Realizations by stochastic finite automata. *Journal of Computer and System Sciences* 5, 26–40.

Cooperman, R., Petrunina, A. E. (2026). The Qp-Foliation Interference Theorem. White paper. DOI: 10.XXXX/XXXXX.

Fliess, M. (1974). Matrices de Hankel. *Journal de Mathématiques Pures et Appliquées* 53, 197–224.

Ireland, K., Rosen, M. (1990). *A Classical Introduction to Modern Number Theory*, 2nd ed. Springer.

Pohlig, S., Hellman, M. (1978). An improved algorithm for computing logarithms over GF(p) and its cryptographic significance. *IEEE Transactions on Information Theory* 24(1), 106–110.

Licence

The text of this paper and the accompanying validation code are released under the following license.


```
MIT License

Copyright (c) 2026 Roger Cooperman and A.E. Petrunina V.V.

Permission is hereby granted, free of charge, to any person obtaining a copy of this software
and associated documentation files (the "Software"), to deal in the Software without
restriction, including without limitation the rights to use, copy, modify, merge, publish,
distribute, sublicense, and/or sell copies of the Software, and to permit persons to whom the
Software is furnished to do so, subject to the following conditions:

The above copyright notice and this permission notice shall be included in all copies or
substantial portions of the Software.

THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING
BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE AND
```

```
NONINFRINGEMENT. IN NO EVENT SHALL THE AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM,
DAMAGES OR OTHER LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM,
OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE SOFTWARE.
```